Follow-Up of MiMeS Magnetic O Star Candidates

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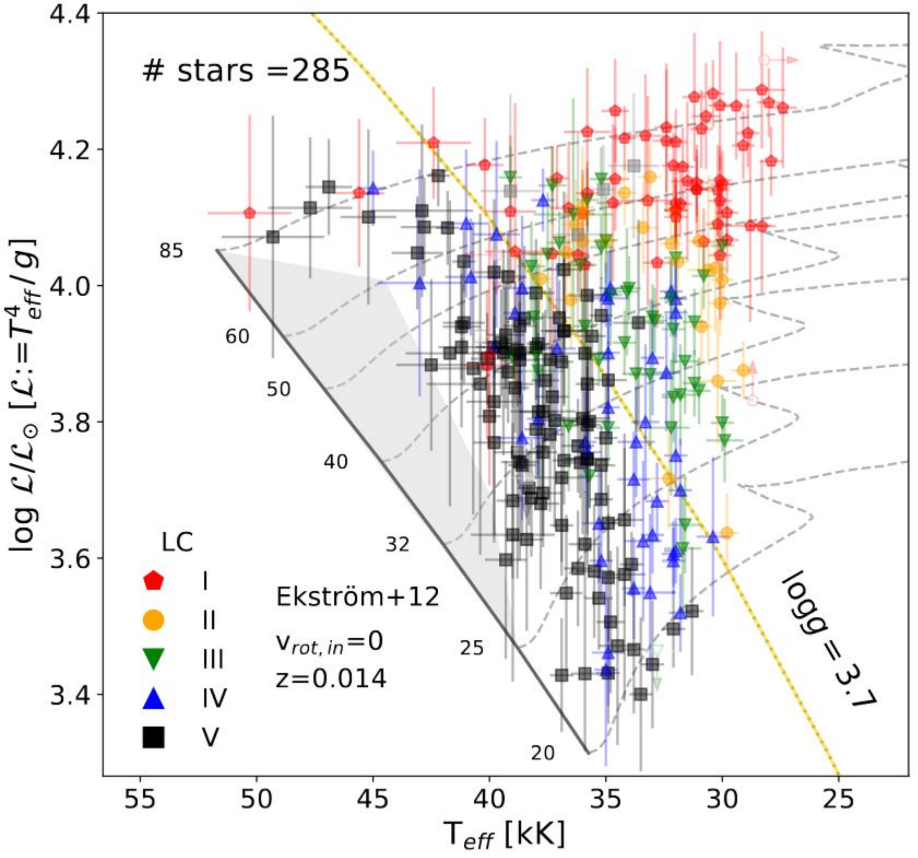
HD 148937

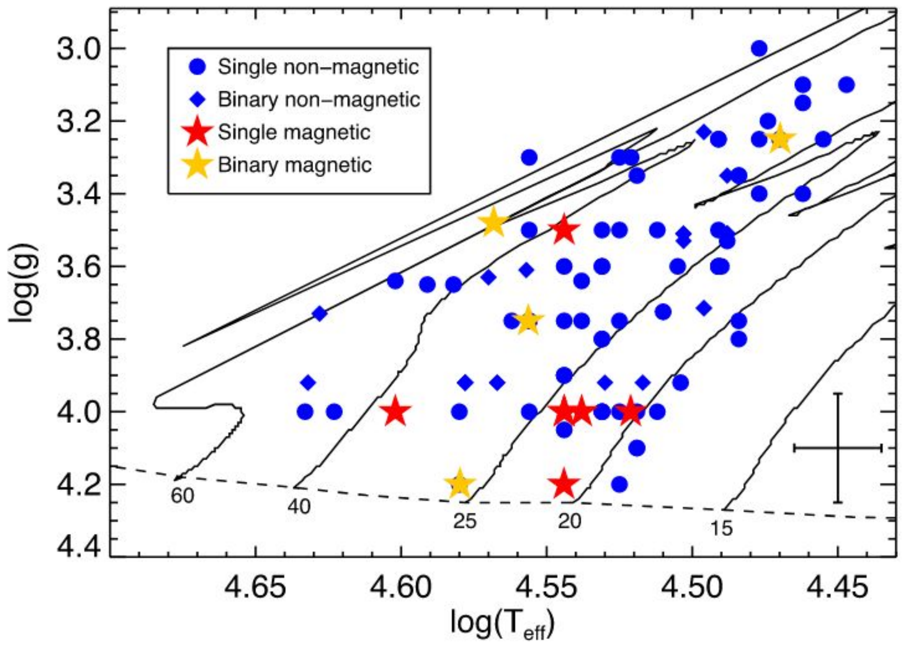
Credit: ESO/VPHAS+

Queen's University

Introduction: MiMeS O-type Stars

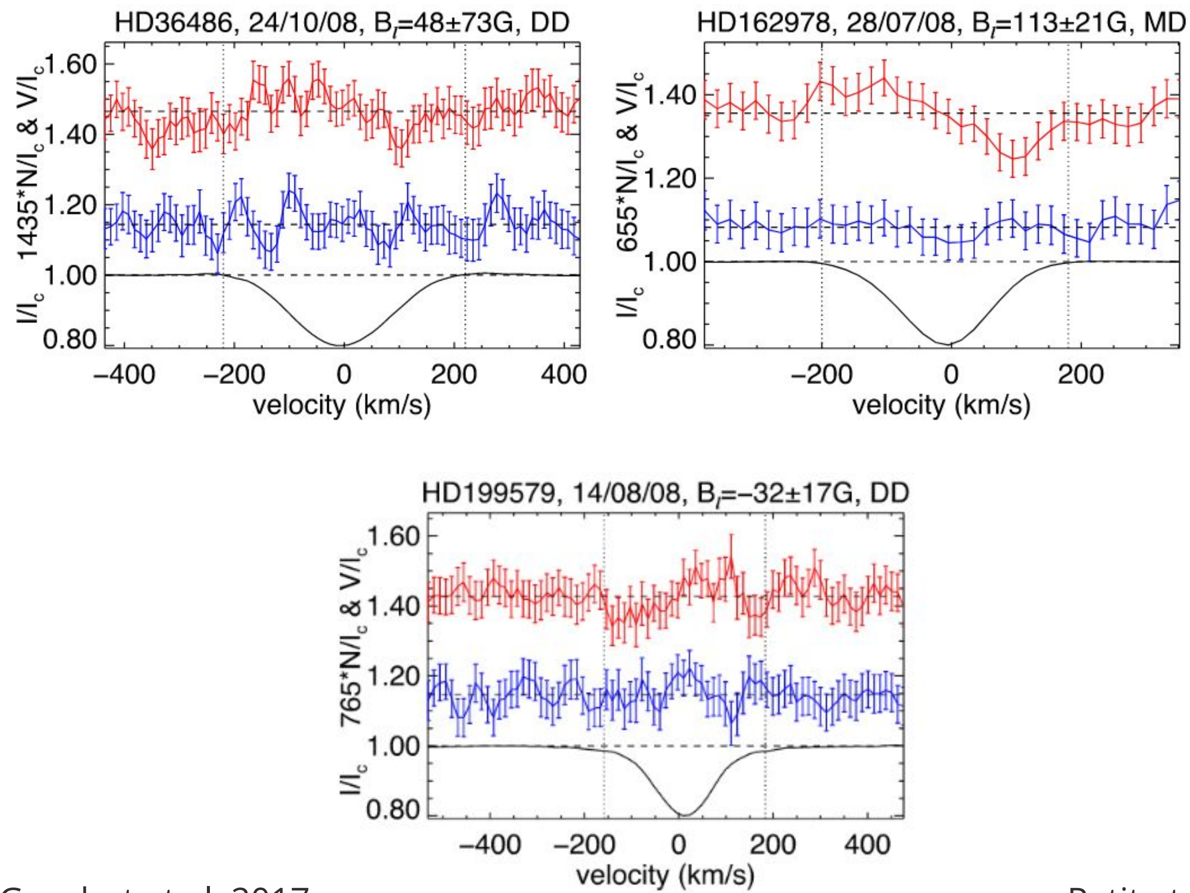


Holgado et al. 2022

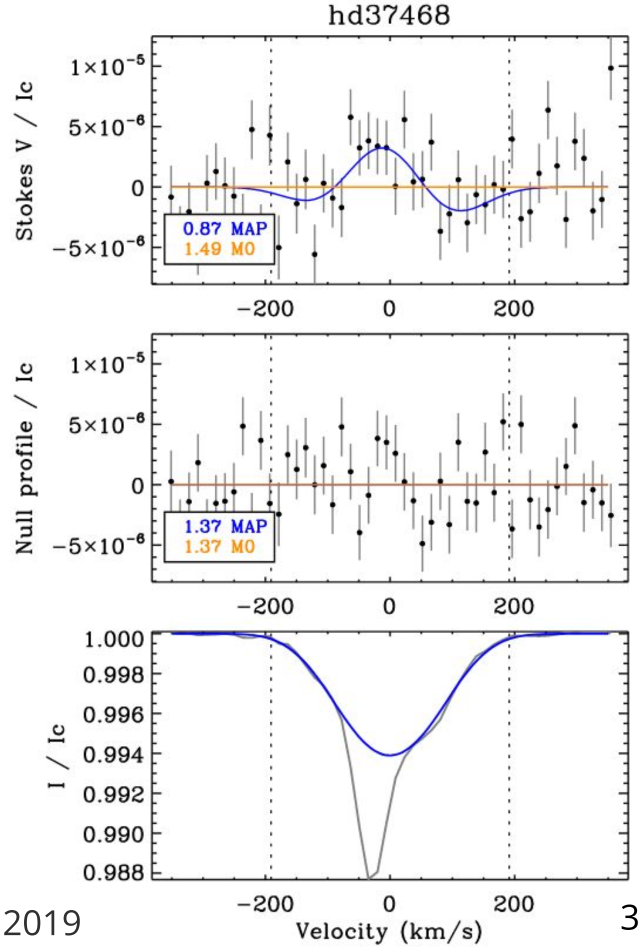


Grunhut et al. 2017

Introduction: MiMeS Magnetic O Star Candidates



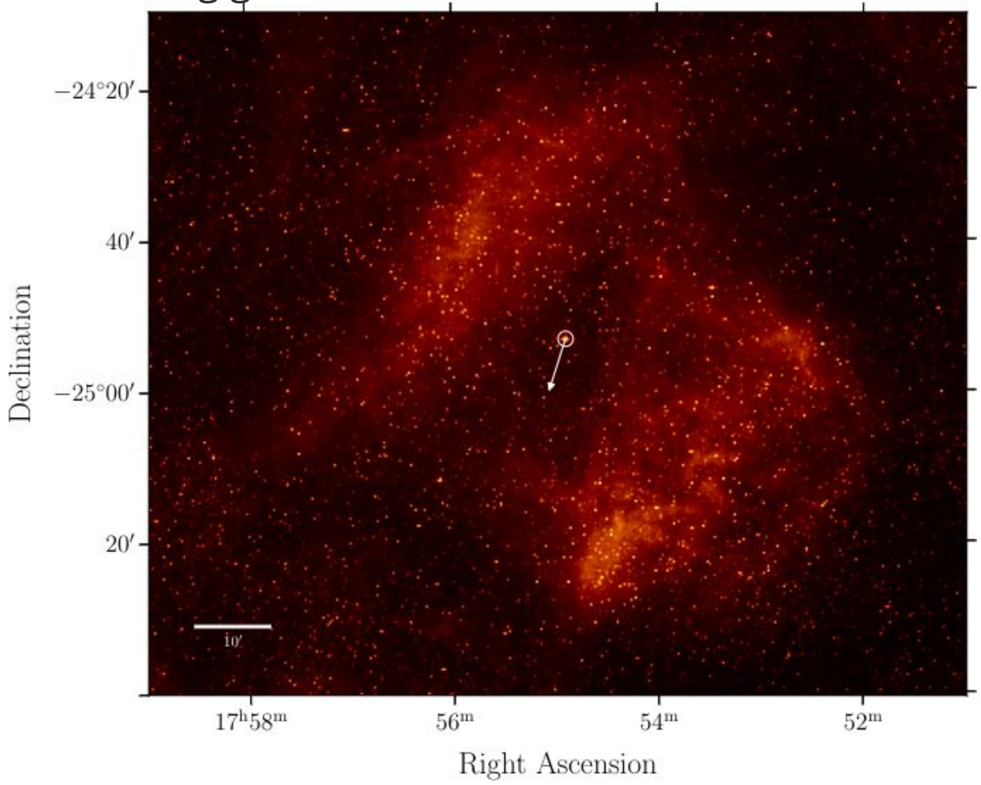
Grunhut et al. 2017

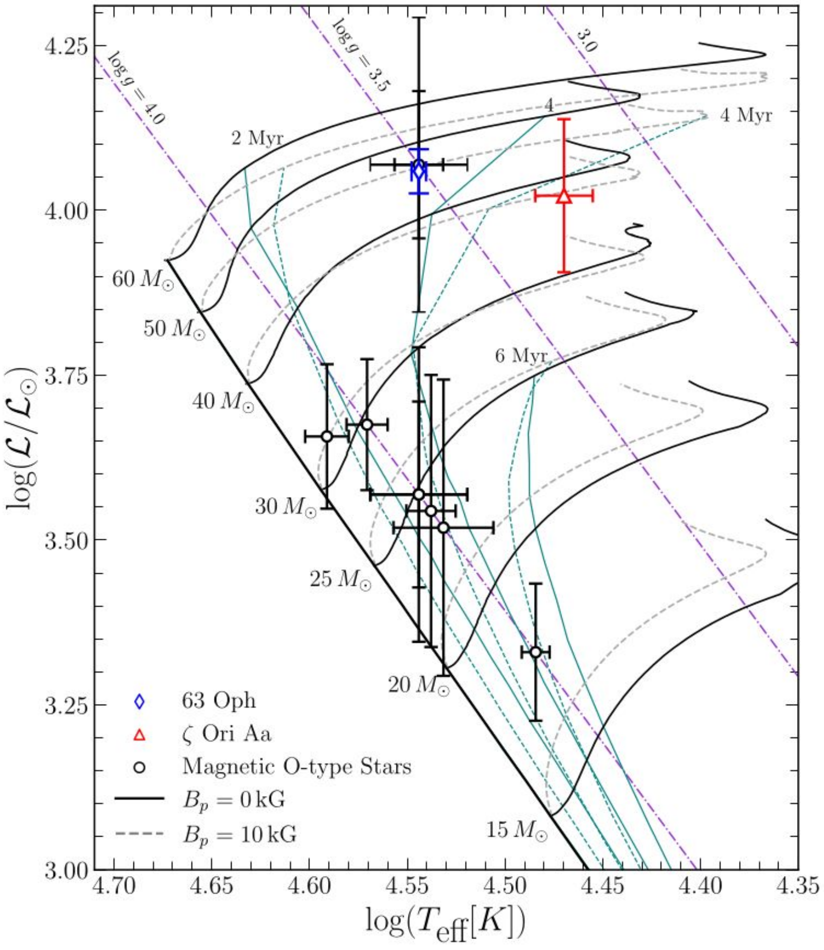


Petit et al. 2019

63 Oph: Introduction

- O8 II bright giant
- Teff = ~ 35 000 K
- Rspec ~ 12 R⊙
- log g ~ 3.5
- Vsini ~ 48 km/s
- Vmac ~ 100 km/s
- Sgr OB1 Association
- d = ~1100 pc

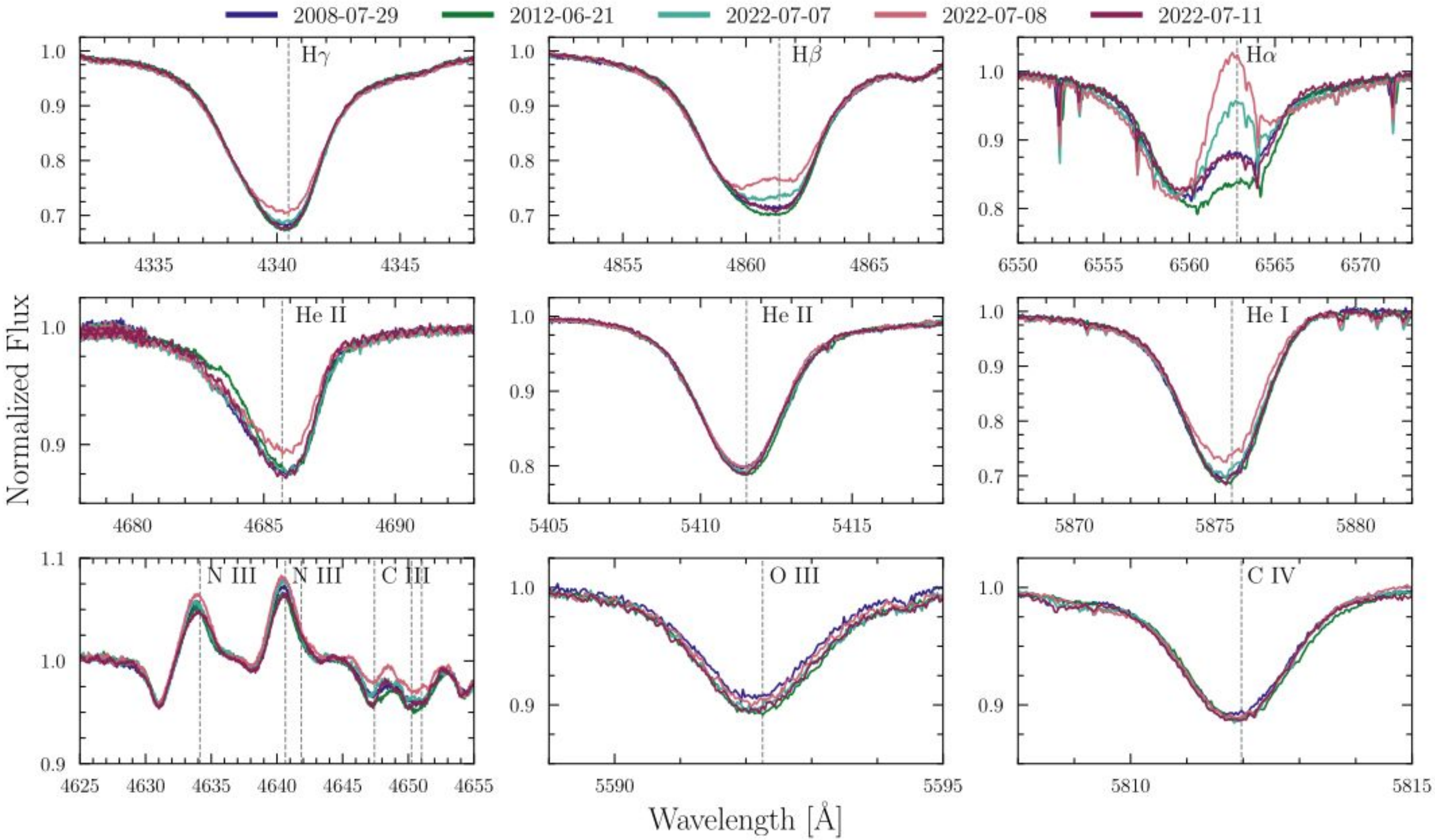




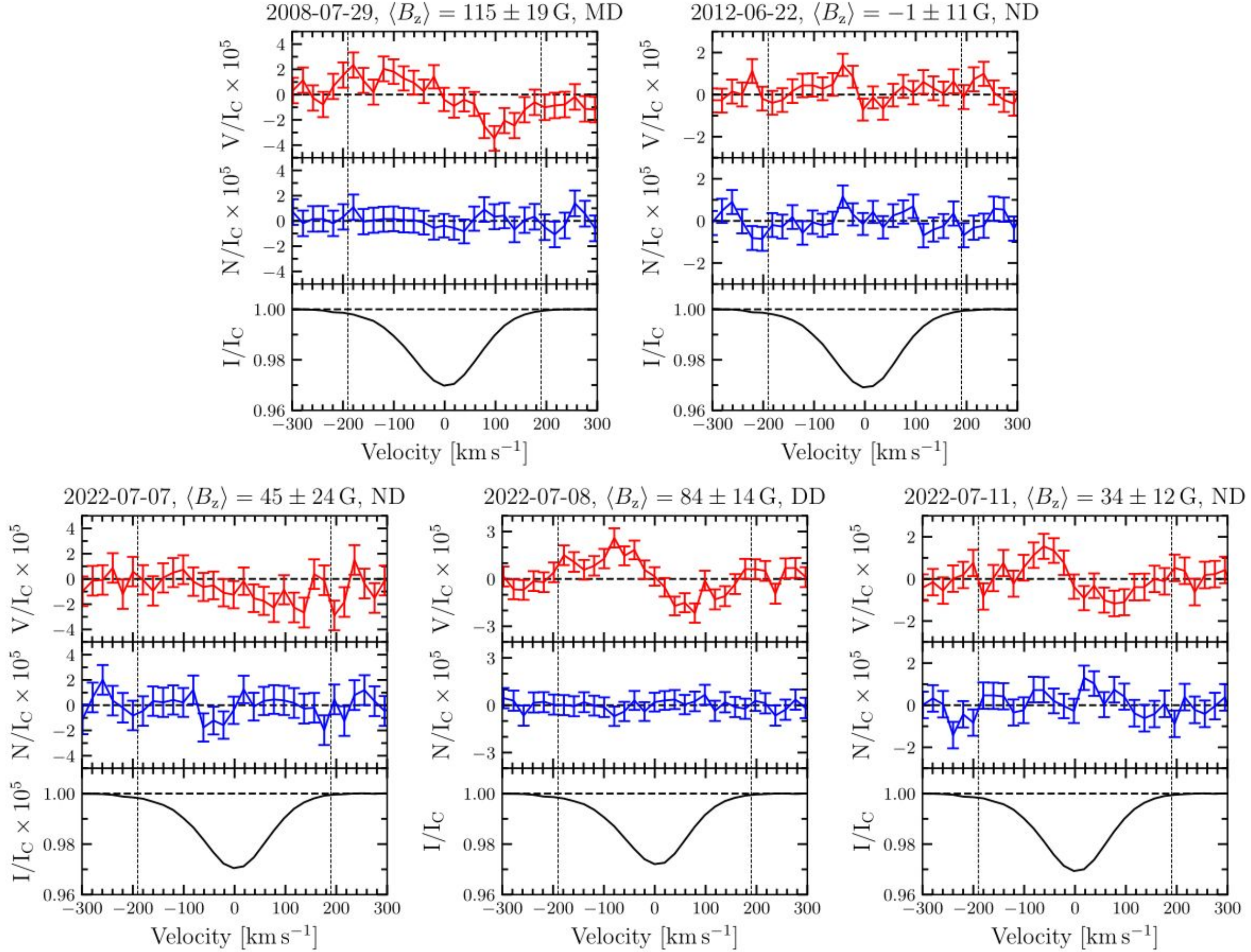
Evolutionary tracks: Keszthelyi et al. 2022

63 Oph: Spectral Variability

35 Spectra from ESPaDOnS, FIES, HERMES and FEROS.



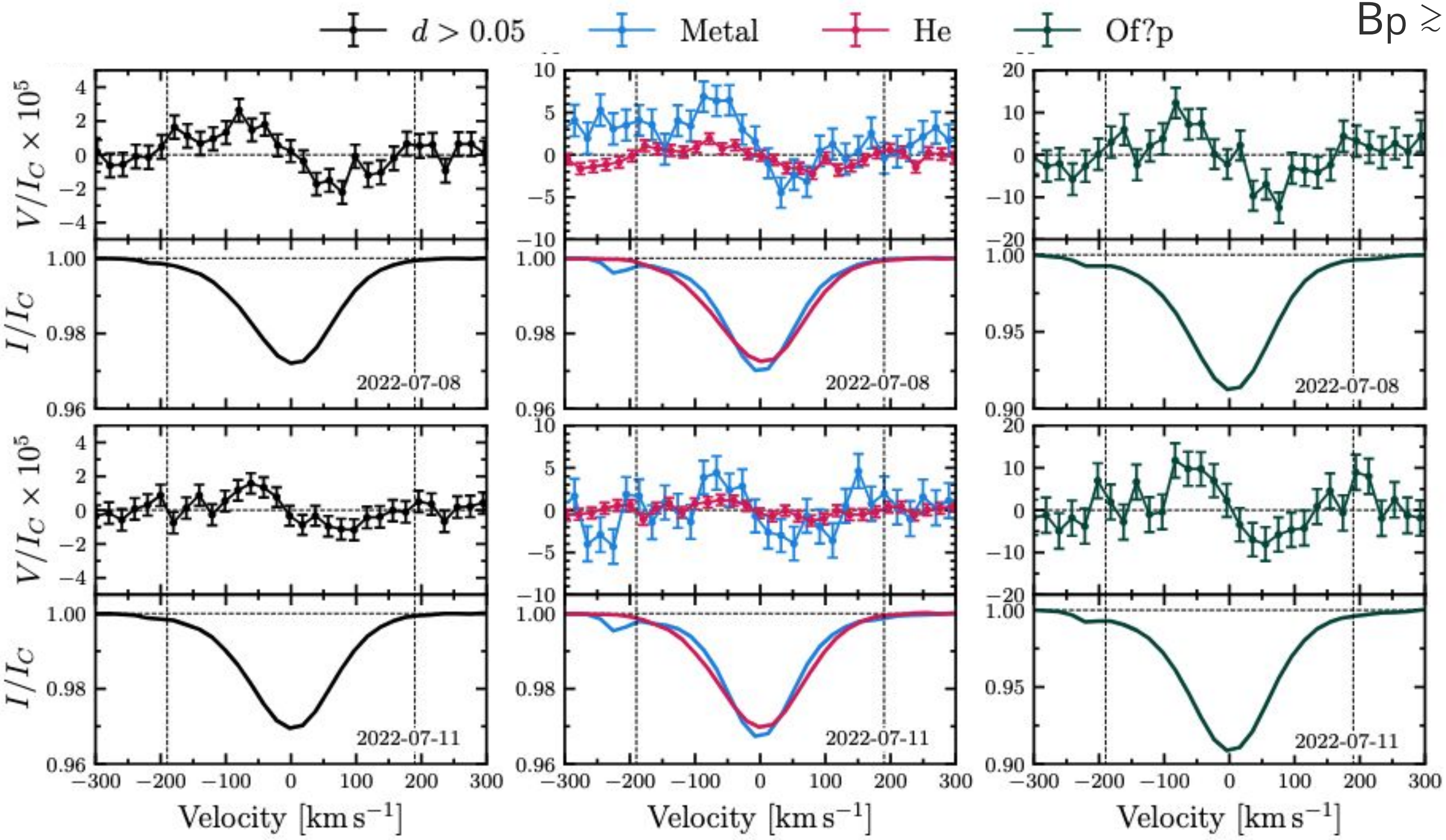
63 Oph: Magnetic Analysis



63 Oph: Magnetic Analysis

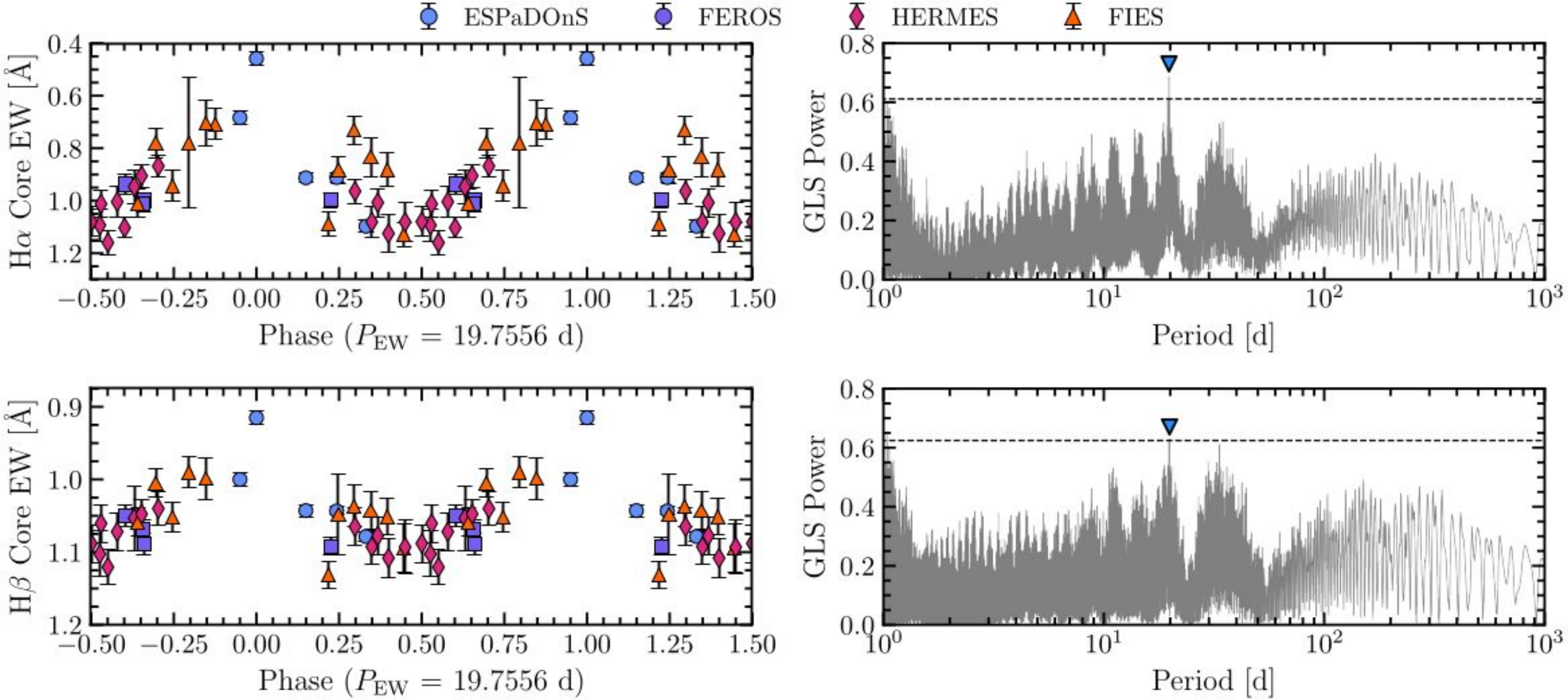
Bz ~ 100 - 200 G

Bp ≳ 300 G

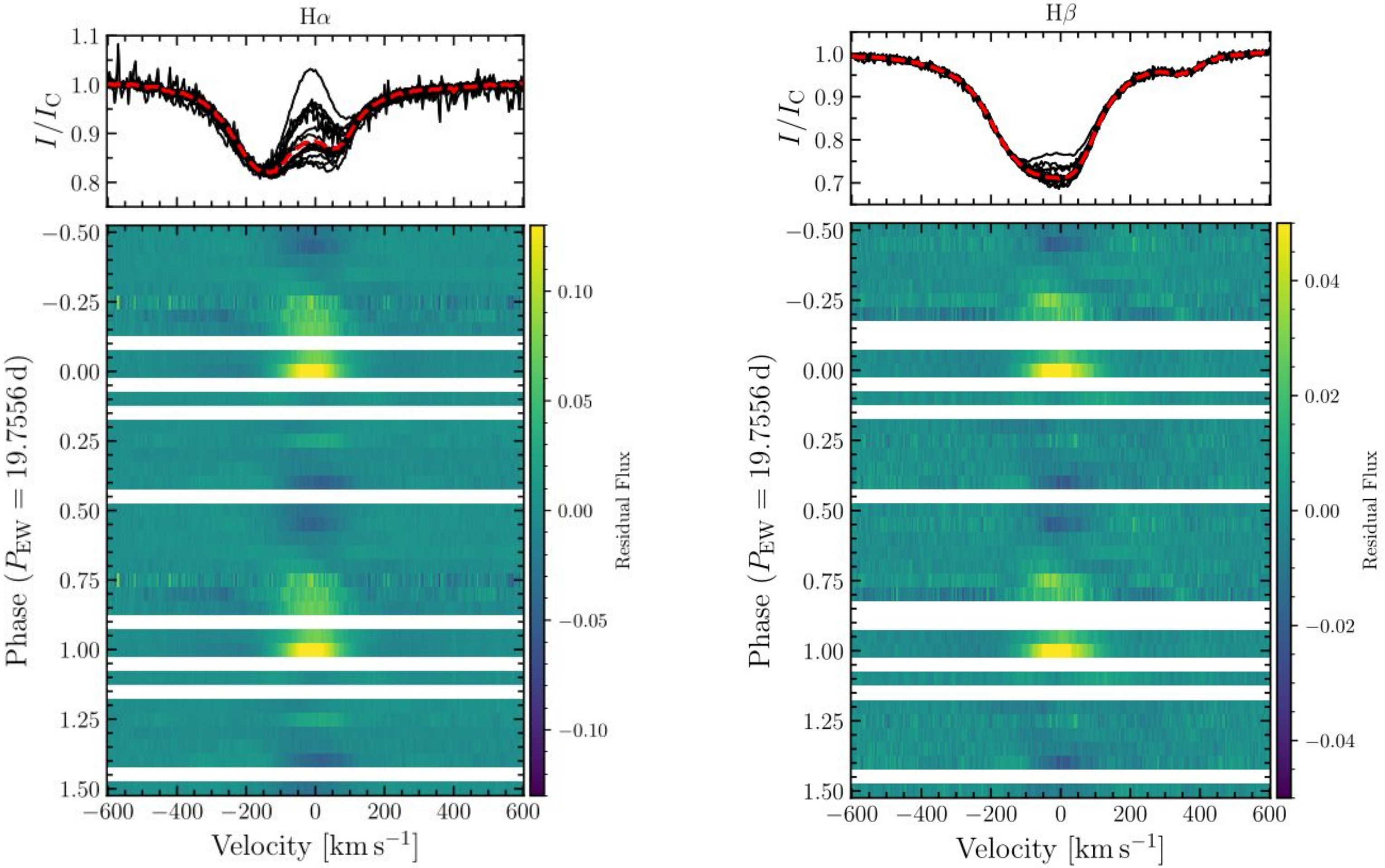


63 Oph: Balmer Line Variability

- ~ 19.8 d period in EWs.
- In tension with maximum rotation period of ~ 13 d given by vsini and Rspec.
- Attribute to rotational modulation of magnetosphere.

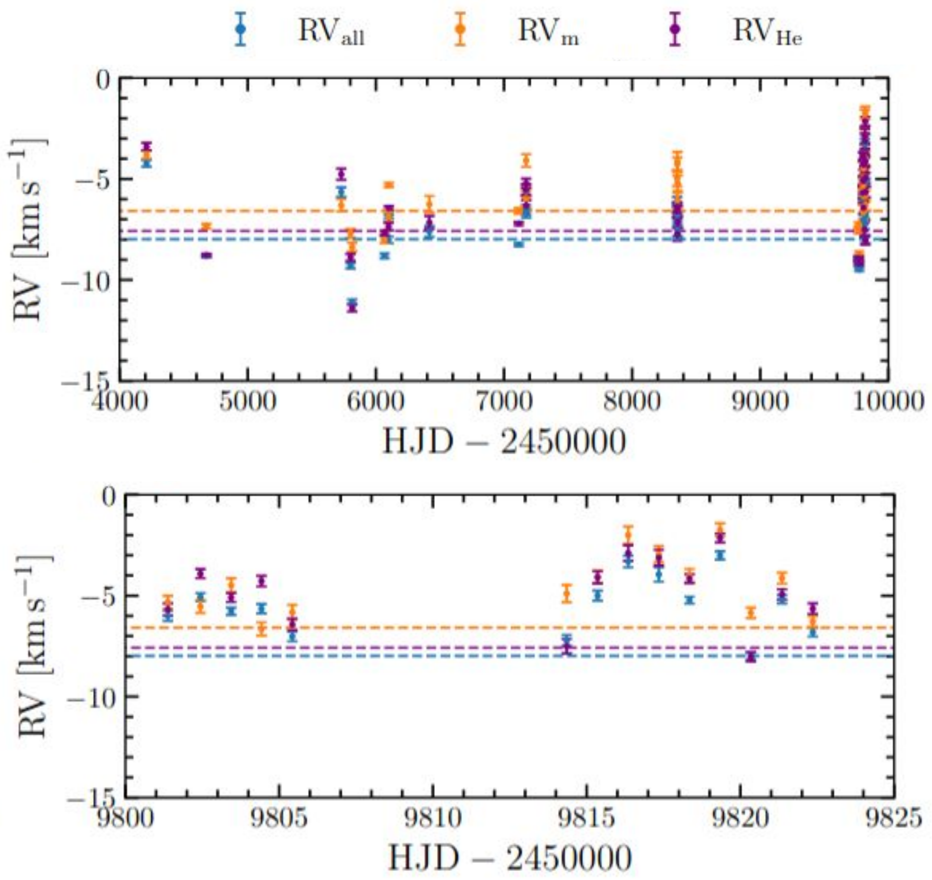


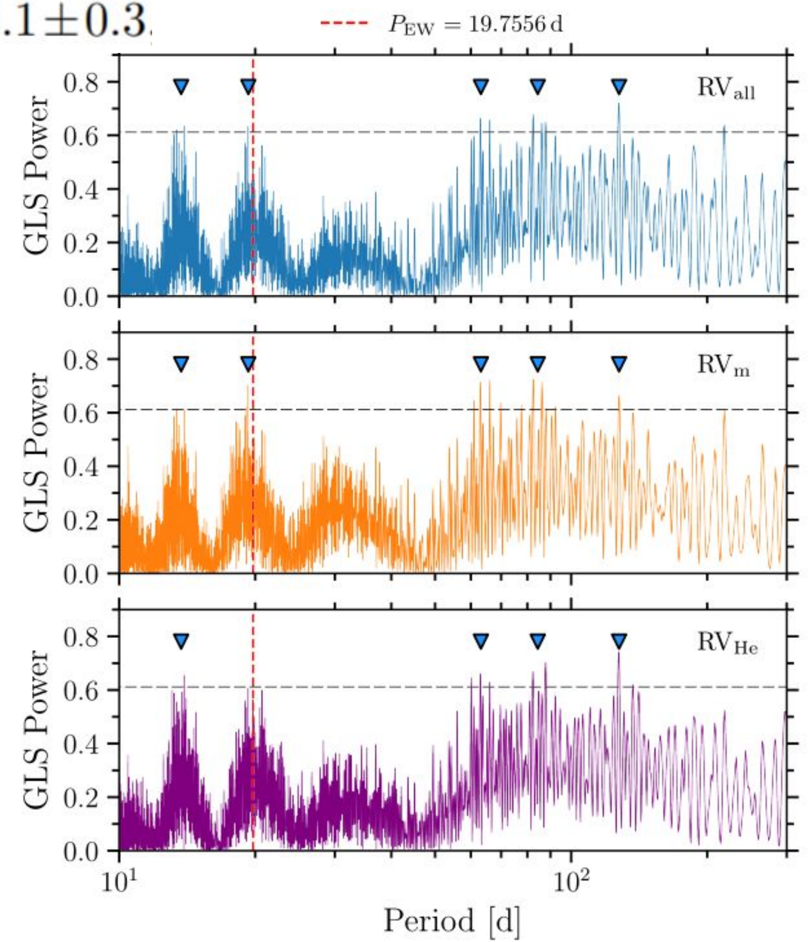
63 Oph: Balmer Line Variability



63 Oph: Radial Velocities

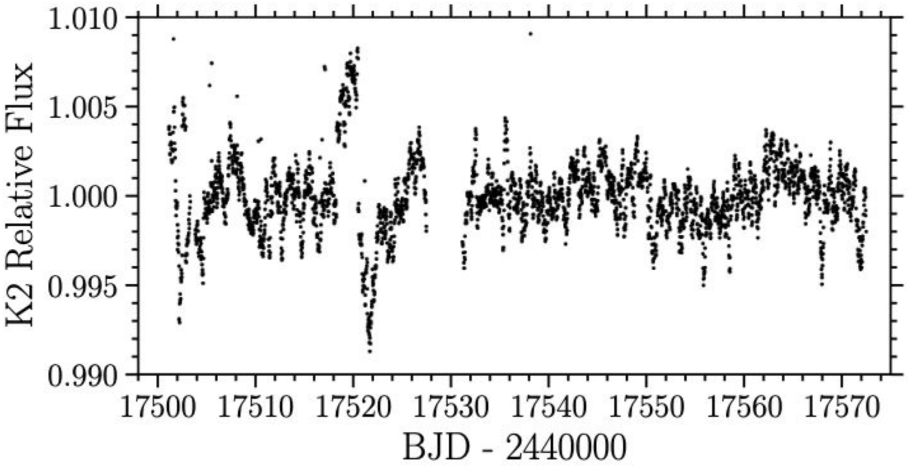
$\Delta RV_{pp,all} = 8.1 \pm 0.3$

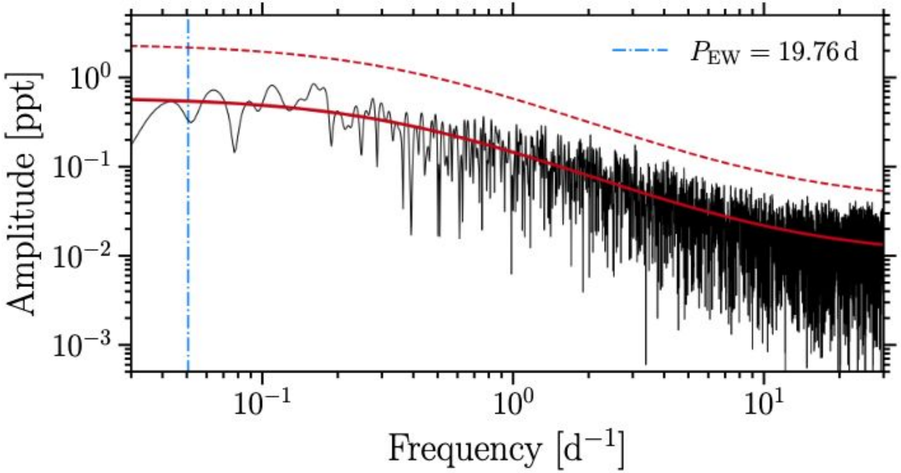


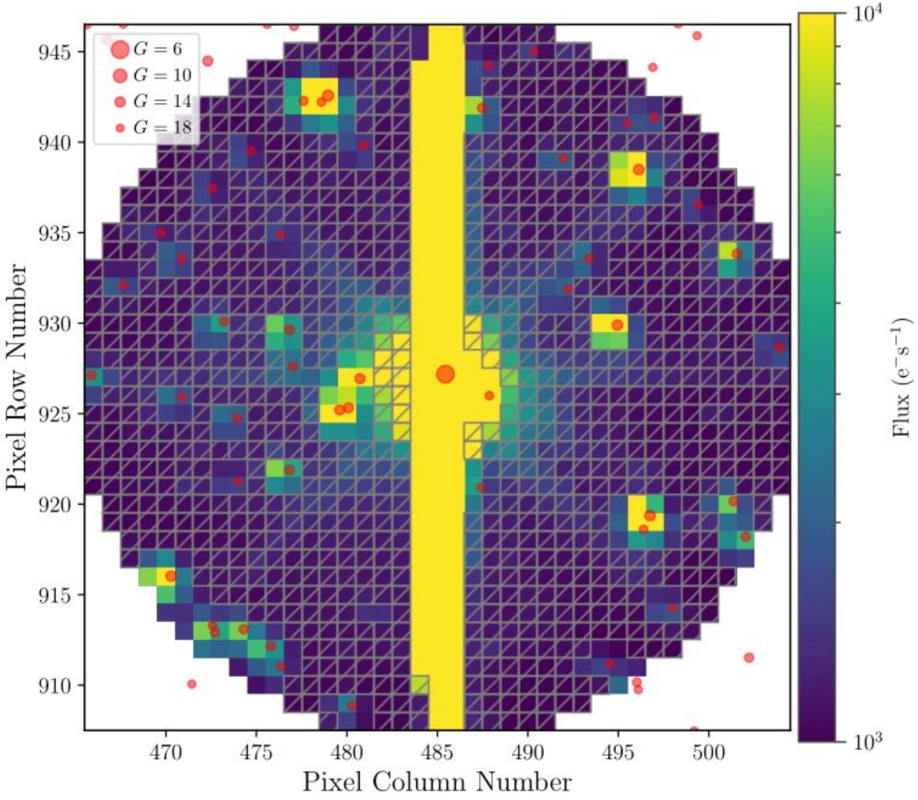


63 Oph: K2 Photometry

Analyzed archival Hipparcos, ASAS-SN and K2 photometry.



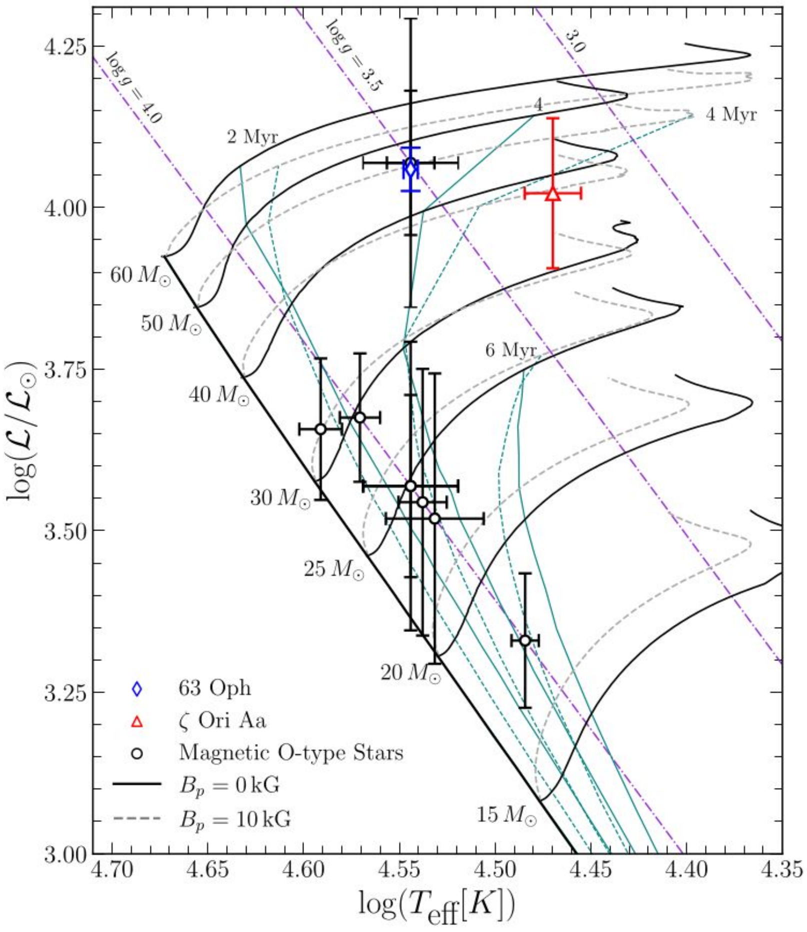




Extraction with halophot (White et al. 2017)

63 Oph: Summary

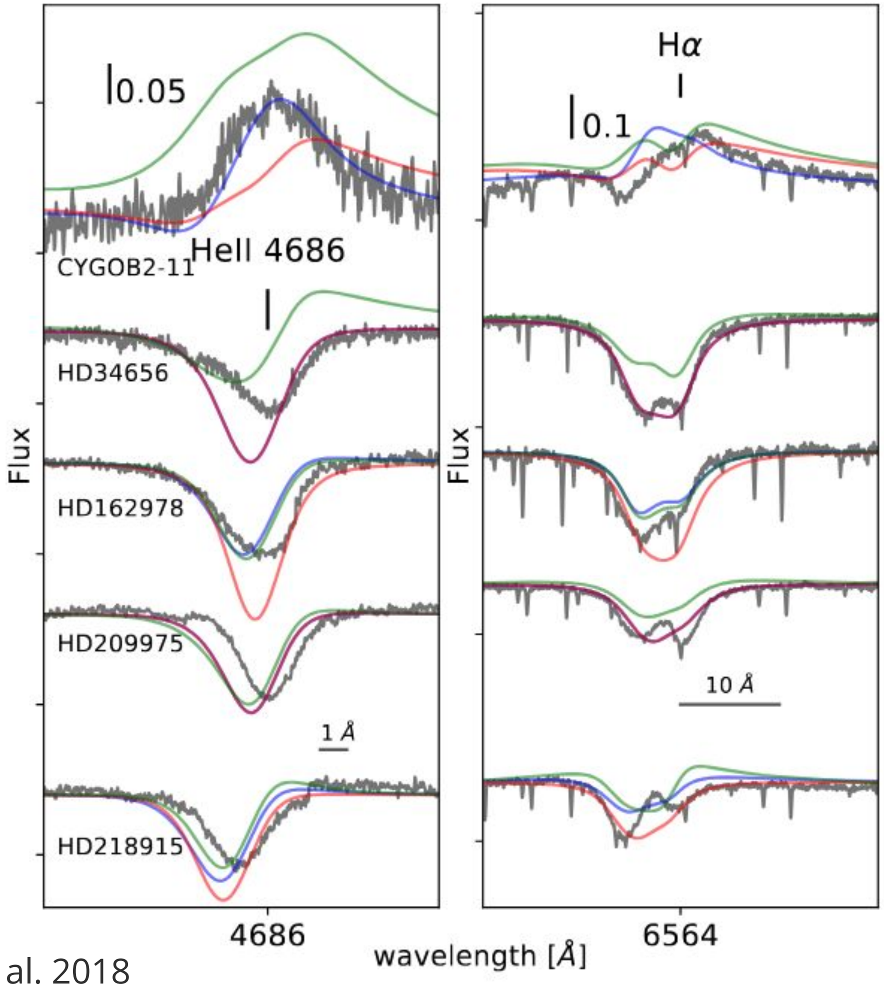
- One MD and one DD, both Bz detected at ~6σ.
- Interpret ~19.8 d EW period as rotational modulation under ORM.
- No clear evidence of binarity from RVs or photometry.
- 63 Oph may be an example of a rare transitional object between the strongly magnetic Of?p stars and ζ Ori A type supergiants.
- Additional observations needed to constraint magnetic geometry and confirm the rotation period.
- Associated nebula should be studied to help test merger hypothesis.



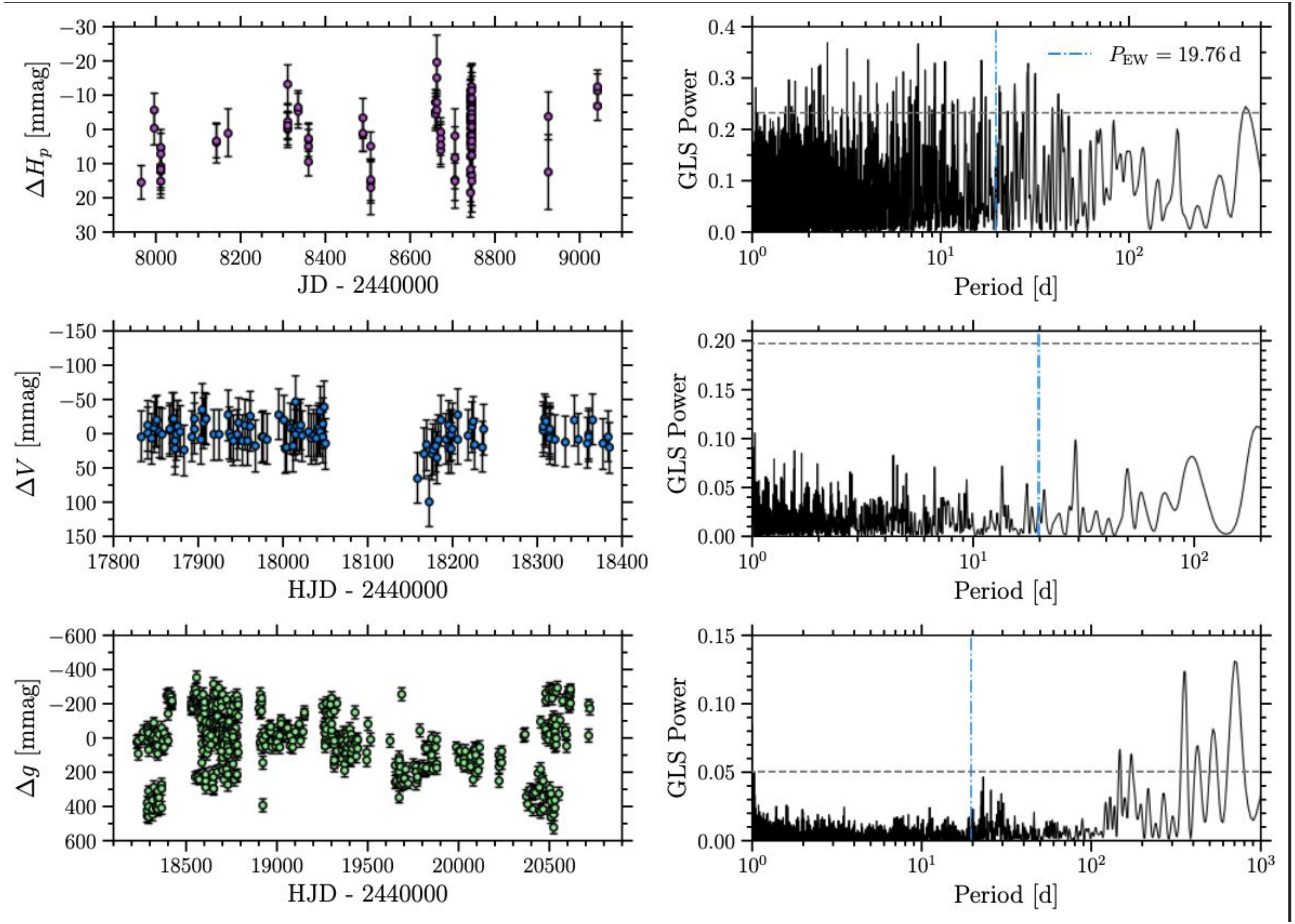
Extra Slides

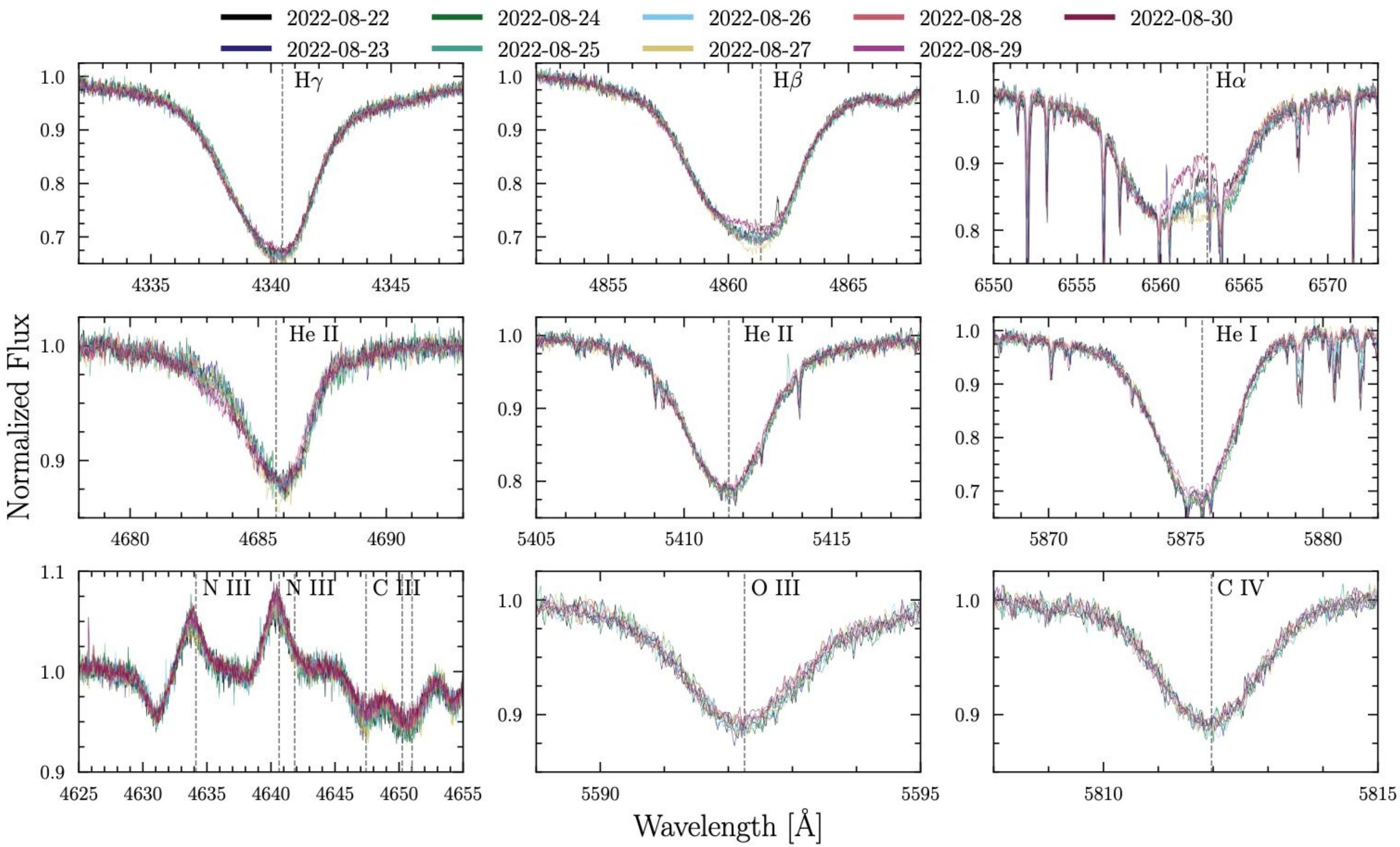
Q3 Flag Stars

- Holgado et al. 2018 Q3 flag - FASTWIND models unable to simultaneously reproduce both Ha and He II 4686 diagnostic wind lines
- 58 IACOB Q3 stars with LC III-I.
- 12 observed by MiMeS.
- 8 have well constrained magnetic fields (Bp < 300 G for (bright) giant, Bp < 150 G for supergiant).
- 2 are detections (63 Oph and ζ Ori A).



Holgado et al. 2018


ΔH_p [mmag]
JD - 2440000
GLS Power
Period [d]
$P_{\mathrm{EW}} = 19.76\,\mathrm{d}$
ΔV [mmag]
HJD - 2440000
GLS Power
Period [d]
Δg [mmag]
HJD - 2440000
GLS Power
Period [d]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2022-08-30
Hγ
Hβ
Hα
He II
He II
He I
N III
N III
C III
O III
C IV
Normalized Flux
Wavelength [Å]